Contents

1 Topological Space

1.1 Basic definition

Definition 1.1.1

Let X be a non-empty set. A set $\mathcal{T}$ of subsets of X is said to be a ***topology*** on X if

(i) X and the empty set, $\varnothing$, belong to $\mathcal{T}$,

(ii) the union of any (finite or infinite) number of sets in $\mathcal{T}$ belongs to $\mathcal{T}$, and

(iii) the intersection of any two sets in $\mathcal{T}$ belongs to $\mathcal{T}$.

The pair $(X, \mathcal{T})$ is called a ***topological space***.

Definition 1.1.2

Let $(X, \mathcal{T})$ be any topological space. Then the members of $\mathcal{T}$ are said to be ***open sets***.

Proposition 1.1.3

If $(X, \mathcal{T})$ is any topological space, then

(i) X and $\varnothing$ are open sets,

(ii) the union of any (finite or infinite) number of open sets is an open set,

(iii) the intersection of any finite number of open sets is an open set.

Definition 1.1.4

Let $(X, \mathcal{T})$ be a topological space. A subset S of X is said to be a ***closed set*** in $(X, \mathcal{T})$ if its complement in X, namely $X \setminus S$, is open in $(X, \mathcal{T})$.

Proposition 1.1.5

If $(X, \mathcal{T})$ is any topological space, then

(i) $\varnothing$ and X are closed sets,

(ii) the intersection of any (finite or infinite) number of closed sets is a

closed set and
(iii) the union of any finite number of closed sets is a closed set.

Definition 1.1.6

A subset $\mathcal{S}$ of a topological space $(X, \mathcal{T})$ is said to be ***clopen*** if it is both open and closed in $(X, \mathcal{T})$.

1.2 Basis for a Topology

Definition 1.2.1

Let $(X, \mathcal{T})$ be a topological space. A collection $\mathcal{B}$ of open subsets of X is said to be a ***basis*** for the topology $\mathcal{T}$ if every open set is a union of members of $\mathcal{B}$.

Remark.
拓扑基不是唯一的，根据定义，一旦一个集族 $\mathcal{B}$ 成为了一个拓扑基，我们无论往 $\mathcal{B}$ 添加多少开集，它还是这个拓扑的基. 因此直觉上，拓扑的基是越小越好用的.

Note.
也就是说，拓扑基是通过任意并运算生成拓扑的对象.

Proposition 1.2.2

Let X be a non-empty set and let $\mathcal{B}$ be a collection of subsets of X. Then $\mathcal{B}$ is a basis for a topology on X if and only if $\mathcal{B}$ has the following properties:
(a) $X = \bigcup_{B \in \mathcal{B}} B$, and
(b) for any $B_1, B_2 \in \mathcal{B}$, the set $B_1 \cap B_2$ is a union of members of $\mathcal{B}$.

Remark.
这里不谈与 X 上的任何一个具体的拓扑的关系，只谈一个集族什么时候能够生成一个拓扑.

Note.
这个命题刻画了一个集族距离成为一组基有多远. 即一个集族需要什么额外条件, 才能通过任意并这个运算生成出一个拓扑. 命题中的条件 (a) 无非对应于 Definition 1.1.1 (i). 而通过基任意并生成的这个操作, 天然就是为了 Definition 1.1.1 (ii) 而建立的. 因此 (b) 就是为了满足1.1.1 (iii) 而提出的条件. 我们发现生成拓扑对于有限交封闭的要求, 完全地变成了对于基的有限交的封闭的要求.

Proof. If $\mathcal{B}$ is a basis for a topology on X, say $\mathcal{T}_{\mathcal{B}}$. Then $X \in \mathcal{T}$ means that there is a collection of sets in $\mathcal{B}$,say $\mathcal{B}_1$, such that

$$X = \bigcup_{B \in \mathcal{B}_i} B \subseteq \bigcup_{B \in \mathcal{B}} B$$

And since every sets in $\mathcal{B}$ is a subset of X, it follows that (a) holds. To show (b), we only need to see that $\mathcal{B} \subseteq \mathcal{T}_{\mathcal{B}}$. Since $\mathcal{T}$ is closed under finite intersection, we have $B_1 \cap B_2 \in \mathcal{T}_{\mathcal{B}}$.

Then conversely , if $\mathcal{B}$ has the properties (a), (b), we only show that the collection $\langle \mathcal{B} \rangle$ of union of the members in $\mathcal{B}$, satisfies Definition 1.1.1 (iii). We take $A_1, A_2 \in \langle \mathcal{B} \rangle$, and suppose that

$$A_i = \bigcup_{j \in I_i} B_j, \quad B_j \in \mathcal{B}, \quad I_i \text{ is a index set }, i = 1, 2$$

Then

$$\begin{aligned}
A_1 \cap A_2 &= \left(\bigcup_{j_1 \in I_1} B_{j_1} \right) \cap \left(\bigcup_{j_2 \in I_2} B_{j_2} \right) \\
&= \bigcup_{j_2 \in I_2} \left(\left(\bigcup_{j_1 \in I_1} B_{j_1} \right) \cap B_{j_2} \right) \\
&= \bigcup_{j_2 \in I_2} \left(\bigcup_{j_1 \in I_1} (B_{j_1} \cap B_{j_2}) \right) \\
&= \bigcup_{j_1 \in I_1, j_2 \in I_2} B_{j_1} \cap B_{j_2}
\end{aligned}$$

Since every $B_{j_1} \cap B_{j_2}$ is a union of members of $\mathcal{B}$, and $A_1 \cap A_2$ is a union of such $B_{j_1} \cap B_{j_2}$, then $A_1 \cap A_2$ is a union of members of $\mathcal{B}$. □

Proposition 1.2.3

Let $(X, \mathcal{T})$ be a topological space. A family $\mathcal{B}$ of open subsets of X is a basis for $\mathcal{T}$ if and only if for any point x belonging to any open set U, there is a $B \in \mathcal{B}$ such that $x \in B \subseteq U$.

Remark.
表述 "there is a $B \in \mathcal{B}$ such that $x \in B \subseteq U$ " 就是在说 "U 可以由 $\mathcal{B}$ 通过 '并' 生成" 的微观讲法. 带入这句话之下, 这个命题几乎就是一句 "废话". 同样的道理也出现在 Proposition 1.2.4

Note.
想要看 $\mathcal{B}$ 是否作为基生成了拓扑 $\mathcal{T}$. 我们不用谈 $\mathcal{B}$ 中的元素如何如何 "并" 出每一个开集 U. 而是只需要 $\mathcal{B}$ 对于开集是足够 "精细" 的, 以至于对于任意的开集 U, 我们可以用 $\mathcal{B}$ 里面的元素在 U 上 "框出" 它的任意一个点.

Proof. If $\mathcal{B}$ is a basis for $\mathcal{T}$. Then we can write U as

$$U = \bigcup_{i \in I} B_i$$

Then for any $x \in U$, there exists $i_0 \in I$ such that $x \in B_{i_0} \subseteq U$.

Now we show the converse. First we show that $\mathcal{B}$ is a basis . We use Proposition 1.2.2 to verify this. Since X is a open set , for any $x \in X$, there is a $B_x \in \mathcal{B}$, such that $x \in B_x \subseteq X$, we get

$$X = \bigcup_{x \in X} B_x \subseteq \bigcup_{B \in \mathcal{B}} B \subseteq X$$

Thus

$$X = \bigcup_{B \in \mathcal{B}} B$$

Take any $B_1, B_2 \in \mathcal{B}$, we know $B_1 \cap B_2$ is again an open set. Then similarly, we get

$$B_1 \cap B_2 = \bigcup_{x \in B_1 \cap B_2} B_x$$

is a union of members in $\mathcal{B}$. Thus $\mathcal{B}$ is actually a basis. Finally , we only need to show that $\mathcal{B}$ generates $\mathcal{T}$, that is $\langle \mathcal{B} \rangle = \mathcal{T}$. $\langle \mathcal{B} \rangle$ is the collection of unions of members in $\mathcal{B}$. $\langle\mathcal{B}\rangle \subseteq \mathcal{T}$ is obvious. $\mathcal{T} \subseteq \langle\mathcal{B}\rangle$ concludes from

$$U = \bigcup_{x\in U} B_x \in \langle\mathcal{B}\rangle$$

□

Proposition 1.2.4

Let $\mathcal{B}$ be a basis for a topology $\mathcal{T}$ on a set X. Then a subset U of X is open if and only if for each $x \in U$ there exists a $B \in \mathcal{B}$ such that $x \in B \subseteq U$.

Note.
把基 $\mathcal{B}$ 看成是 $\mathcal{T}$ 中代表着“足够精细”的某一批. 验证 U 是否是开集, 只需要看 U 是否可以被这一小批集合“概括”.

Proof. $\implies$ is immediately from Proposition 1.2.3. Now suppose that for each $x \in U$, ther exists a $B_x \in \mathcal{B}$ such that $x \in B \subseteq U$, then

$$U = \bigcup_{x\in U} B_x \in \langle\mathcal{B}\rangle = \mathcal{T}$$

□

Proposition 1.2.5

Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be bases for topologies $\mathcal{T}_1$ and $\mathcal{T}_2$, respectively, on a non-empty set X. Then $\mathcal{T}_1 = \mathcal{T}_2$ if and only if

1. for each $B \in \mathcal{B}_1$ and each $x \in B$, there exists a $B' \in \mathcal{B}_2$ such that $x \in B' \subseteq B$, and
2. for each $B \in \mathcal{B}_2$ and each $x \in B$, there exists a $B' \in \mathcal{B}_1$ such that $x \in B' \subseteq B$.

Proof sketch.
考虑这句话: 表述 “there is a $B \in \mathcal{B}$ such that $x \in B \subseteq U$ ” 就是在说 “U 可以由 $\mathcal{B}$ 通过 ‘并’ 生成”. 1. 是说 $\mathcal{B}_1$ 中任意集合可以由 $\mathcal{B}_2$“并” 出来. 自然地 $\mathcal{T}_1$ 中任意集合也被 $\mathcal{B}_2$“并出来”. 因此 1. 就是 $\mathcal{T}_1 \subseteq \mathcal{T}_2$. 同样地, 2. 就是 $\mathcal{T}_2 \subseteq \mathcal{T}_1$.

Lemma 1.2.6

Let $\mathcal{B}$ and $\mathcal{B}'$ be bases for the topologies $\mathcal{T}$ and $\mathcal{T}'$, respectively, on X. Then the following are equivalent:

1. $\mathcal{T}'$ is finer than $\mathcal{T}$.
2. For each $x \in X$ and each basis element $B \in \mathcal{B}$ containing x, there is a basis element $B' \in \mathcal{B}'$ such that $x \in B' \subset B$.

Proof sketch.
同样的, 2. 就是在说每个 B 可以由一堆 B' 并出来. 那么显然每个 $U \in \mathcal{T}$ 可以由一堆 B' 并出来, 即 $U' \in \mathcal{T}$.

1.3 Closed Sets and Limit Points

Definition 1.3.1

A subset A of a topological space X is said to be ***closed*** if the set $X - A$ is open.

1.3.1 Closure, Interior

Definition 1.3.2

Given a subset A of a topological space X

- The ***interior*** of A , denoted by A° or by $\operatorname{Int} A$, is defined as the union of all open sets contained in A.
- The ***closure*** of A , denoted by $\bar{A}$ or by $\operatorname{Cl} A$, is defined as the intersection of all closed sets containing A.

Theorem 1.3.3

Let A be a subset of the topological space X.

(a) Then $x \in \bar{A}$ if and only if every open set U containing x intersects A.

(b) Supposing the topology of X is given by a basis, then $x \in \bar{A}$ if and only if every basis element B containing x intersects A.

Proof. (a) Method 1. Note that

$$V \text{ is a closed set containing } A \iff V^c \text{ is an open set such that } V^c \cap A = \varnothing$$

We have

$$
\begin{aligned}
x \notin \bar{A} &\iff x \notin \bigcap_{V \text{ is a closed set containing A}} V \\
&\iff x \in \bigcup_{V \text{ is a closed set containing } A} V^c \\
&\iff x \in \bigcup_{U \text{ is an open set such that } U \cap A = \varnothing} U \\
&\iff \text{there exists open set } U \text{ such that } x \in U, U \cap A = \varnothing
\end{aligned}
$$

Method 2. Suppose that $x \in \bar{A}$, then x is contained in any closed set containing A . If there exists open set U containing x dose not intersects A. Then U^c is a closed set containing A, which implies that $x \in U^c$, a contradiction. Conversely , if every open set containing x intersects A, then every closed set that x dose not lie in cannot contain A. i.e. x lies in every closed set containing A.

(b) If every open set containing x intersects A, then every basis B containing x intersects A since B is also an open set . Conversely, if every basis B containing x intersects A, since every open set U contains a basis contaning x , it intersects with A as well. Then it follows from (a).

□

1.3.2 Limit Points

Definition 1.3.4

If A is a subset of the topological space X and if x is a point of X, we say that x is a ***limit point*** (or "cluster point," or "point of accumulation") of A if every neighborhood of x intersects A in some point ***other than x itself***. Said differently, x is a limit point of A if it belongs to the closure of $A - \{x\}$.

Theorem 1.3.5

Let A be a subset of the topological space X; let A' be the set of all ***limit points*** of A. Then

$$\bar{A} = A \cup A'.$$

Proof sketch.
对于 $\bar{A} \subseteq A \cup A'$, $A \subseteq \bar{A}$ 显然而 $A' \subseteq A$ 是 Theorem 1.3.3的直接推论. $A \cup A' \subseteq \bar{A}$, 则考虑若 x 不在 A 里面, 那么 $x \in A'$ 的定义中的开集一定与 A 相交, 从而有 $x \in \bar{A}$.

1.4 The Subspace Topology

Definition 1.4.1

Let X be a topological space with topology $\mathcal{T}$. If Y is a subset of X, the collection
$$\mathcal{T}_Y = \{Y \cap U : U \in \mathcal{T}\}$$
is a topology on Y , called the ***subspace topology***. With this topology, Y is called ***subspace*** of X; its open sets consist of all intersections of open sets of X with Y.

Lemma 1.4.2

If $\mathcal{B}$ is a basis for the topology of X then the collection
$$\mathcal{B}_Y = \{B \cap Y \mid B \in \mathcal{B}\}$$
is a basis for the subspace topology on Y.

Lemma 1.4.3

Let Y be a subspace of X. If U is open in Y and Y is open in X, then U is open in X.

1.5 The Product Topology

1.5.1 Product of the Two

Theorem 1.5.1

If $\mathcal{B}$ is a ***basis*** for the topology of X and $\mathcal{C}$ is a ***basis*** for the topology of Y, then the collection
$$\mathcal{D} = \{B \times C \mid B \in \mathcal{B} \text{ and } C \in \mathcal{C}\}$$

is a ***basis*** for the topology of $X \times Y$.

Definition 1.5.2

Let $\pi_1 : X \times Y \to X$ be defined by the equation

$$\pi_1(x, y) = x;$$

let $\pi_2 : X \times Y \to Y$ be defined by the equation

$$\pi_2(x, y) = y.$$

The maps π_1 and π_2 are called the ***projections*** of $X \times Y$ onto its first and second factors, respectively.

Theorem 1.5.3

The collection

$$\mathcal{S} = \{\pi_1^{-1}(U) \mid U \text{ open in } X\} \cup \{\pi_2^{-1}(V) \mid V \text{ open in } Y\}$$

is a ***subbasis*** for the ***product topology*** on $X \times Y$.

1.5.2 The General Product

Definition 1.5.4

Let J be an index set. Given a set X, we define a J***-tuple*** of elements of X to be a function $\mathbf{x} : J \to X$. If α is an element of J, we often denote the value of $\mathbf{x}$ at α by x_α rather than $\mathbf{x}(\alpha)$; we call it the αth ***coordinate*** of $\mathbf{x}$. And we often denote the function $\mathbf{x}$ itself by the symbol

$$(x_\alpha)_{\alpha \in J},$$

which is as close as we can come to a "tuple notation" for an arbitrary index set J. We denote the set of all J-tuples of elements of X by X^J.

Definition 1.5.5

Let $\{A_\alpha\}_{\alpha \in J}$ be an indexed family of sets; let $X = \bigcup_{\alpha \in J} A_\alpha$. The ***carte-***

sian product of this indexed family, denoted by

$$\prod_{\alpha \in J} A_\alpha,$$

is defined to be the set of all J-tuples $(x_\alpha)_{\alpha \in J}$ of elements of X such that $x_\alpha \in A_\alpha$ for each $\alpha \in J$. That is, it is the set of all functions

$$\mathbf{x} : J \to \bigcup_{\alpha \in J} A_\alpha$$

such that $\mathbf{x}(\alpha) \in A_\alpha$ for each $\alpha \in J$.

Definition 1.5.6

Let $\{X_\alpha\}_{\alpha \in J}$ be an indexed family of topological spaces. Let us take as a basis for a topology on the product space

$$\prod_{\alpha \in J} X_\alpha$$

the collection of all sets of the form

$$\prod_{\alpha \in J} U_\alpha,$$

where U_α is open in X_α, for each $\alpha \in J$. The topology generated by this basis is called the ***box topology***.

Definition 1.5.7

Let S_β denote the collection

$$S_\beta = \{\pi_\beta^{-1}(U_\beta) \mid U_\beta \text{ open in } X_\beta\},$$

and let $\mathcal{S}$ denote the union of these collections,

$$\mathcal{S} = \bigcup_{\beta \in J} S_\beta.$$

The topology generated by the subbasis $\mathcal{S}$ is called the ***product topology***. In this topology $\prod_{\alpha \in J} X_\alpha$ is called a ***product space***.

Theorem 1.5.8 ▸ Comparison of the *box and product topologies*

The box topology on $\prod X_\alpha$ has as basis all sets of the form $\prod U_\alpha$, where U_α is open in X_α for each α. The product topology on $\prod X_\alpha$ has as basis all sets of the form $\prod U_\alpha$, where U_α is open in X_α for each α and U_α equals X_α except for finitely many values of α.

Theorem 1.5.9

Suppose the topology on each space X_α is given by a basis $\mathcal{B}_\alpha$. The collection of all sets of the form

$$\prod_{\alpha \in J} B_\alpha,$$

where $B_\alpha \in \mathcal{B}_\alpha$ for each α, will serve as a basis for the box topology on $\prod_{\alpha \in J} X_\alpha$. The collection of all sets of the same form, where $B_\alpha \in \mathcal{B}_\alpha$ for finitely many indices α and $B_\alpha = X_\alpha$ for all the remaining indices, will serve as a basis for the product topology $\prod_{\alpha \in J} X_\alpha$.

Theorem 1.5.10

Let A_α be a subspace of X_α, for each $\alpha \in J$. Then $\prod A_\alpha$ is a subspace of $\prod X_\alpha$ if both products are given the box topology, or if both products are given the product topology.

Theorem 1.5.11

If each space X_α is ***Hausdorff space***, then $\prod X_\alpha$ is a ***Hausdorff space*** in both the ***box*** and ***product topologies***.

Theorem 1.5.12

Let $\{X_\alpha\}$ be an indexed family of spaces; let $A_\alpha \subset X_\alpha$ for each α. If $\prod X_\alpha$ is given either the product or the box topology, then

$$\prod \bar{A}_\alpha = \overline{\prod A_\alpha}.$$

Theorem 1.5.13

Let $f : A \to \prod_{\alpha \in J} X_\alpha$ be given by the equation

$$f(a) = (f_\alpha(a))_{\alpha \in J},$$

where $f_\alpha : A \to X_\alpha$ for each α. Let $\prod X_\alpha$ have the product topology. Then the function f is continuous if and only if each function f_α is continuous.